



COMPUTER AWARENESS

- The Boolean expression $AB+AB'+A'C+AC$ is unaffected by the value of the Boolean variable
(a)C (b)A (c)B (d)None
- Suppose the largest n bit number requires, ' d ' digits in decimal representation. Which of the following relations between ' n ' and ' d ' approximately correct?
(a) $d = 2^n$ (b) $n = 2^d$ (c) $d < n \log_{10} 2$ (d) $d > n \log_{10} 2$
- FFFF will be the last memory location in a memory is size
(a)32k (b)16k (c)64k (d)1k
- Which of the following is equivalent to the Boolean expression $(X+Y) \cdot (\bar{X} + \bar{Y})$.
(a)XY (b) $\bar{Y}X$ (c) $\bar{X}Y$ (d) $\bar{X}\bar{Y}$
- 'Floating point representation' is used to represent
(a)Integers (b)Real number
(c)Boolean values (d)Whole numbers
- If a processor clock is rated as 2500 million cycles per second, then its clock period is
(a) 2.50×10^{-10} sec (b) 4.00×10^{-10} sec
(c) 1.00×10^{-10} sec (d)None of these
- The minimum number of NAND gates required for implementing of Boolean expression, $AB + \bar{A}\bar{B} + \bar{A}B\bar{C}$ is
(a)1 (b)0 (c)2 (d)3
- The maximum and minimum value represented in signed 16 bit 2's complement representation are
(a)-16384 and 16383 (b)0 and 32767
(c)-32768 and 32767 (d)0 and 65535
- Write the simplified form of the Boolean expression $(A+C)(A+AD)+AC+C$
(a) $A+C'$ (b) $A+C$ (c) $A'+C$ (d) $A+D$
- If a signal passing through a gate is inhibited by sending a low into one of the inputs, and the output is high, the gate is a(n)
(a)OR (b)NOR (c)NAND (d)AND

ANSWER

1.	C	2.	D	3.	C	4.	A	5.	B
6.	B	7.	B	8.	C	9.	B	10.	c

SOLUTION

- (c) Let $F(A, B, C) = AB + AB' + A'C + AC$
 $F(A, B, C) = A(B + B') + C(A' + A)$
 $= A(1) + C(1) = A + C$
 Since, the minimised Boolean expression does not contain Boolean variable B . Therefore, $F(A, B, C)$ is unaffected by the value of the Boolean variable B .
- (d) n bits binary number requires d -decimal digits
 So,
 $10^d > 2^n$
 Take log on both side
 $\log_{10} 10^d > \log_{10} 2^n$
 $d > n \log_{10} 2$
- (c) FFFF will be the last memory location in a memory size 64 K. The equivalent addressing ranges are 0 to 65535 in decimal and 0 to FFFF in hex.
 Hexadecimal (also base 16) is a positional numeral system with a radix or base of 16. It uses sixteen distinct symbols, most often

the symbols 0 – 9 to represent values zero to nine and A – F to represent values ten to fifteen.

Bytes	Hex	Address range
16	10	0 – F
256	100	0 – FF
1 K	400	0 – 3FF
4 K	1000	0 – FFF
64 K	10000	0 – FFFF

- (a) $(x+y)(x+y')(x'+y)$
 $= [(x+y)(x+y')](x'+y)$
 $= [x + xy' + xy + yy'](x'+y)$
 $= [x + xy' + xy + 0](x'+y)$
 $= [x + x(y' + y)](x'+y)$
 $= [x + x(1)](x'+y)$
 $= [x + x](x'+y)$
 $= x(x'+y)$
 $= xx' + xy$
 $= 0 + xy = xy$
- (b) Floating point representation is used to represent real numbers. This representation does not reserve a specific number of bits for the integer part or the fractional part.
 The floating number representation of a number has two part : the first part represents a signed fixed point number called Mantissa and the second part of designates the position of the decimal point and is called the exponent.
- (b) We know that frequency is defined as the number of cycles in one second.
 Number of cycle in 1sec = 2500 million
 Frequency = 2500MHz
 We know that time period is the inverse of frequency and is defined as the time taken by one cycle.
 $\Rightarrow T = 1/F$
 $= \frac{1}{2500 \times 10^6}$
 $= \frac{100}{2500 \times 10^6 \times 100} = \frac{100}{25 \times 10^{10}}$
 $= \frac{100}{25} \times 10^{-10} = 4 \times 10^{-10}$ sec
- (b) Let $F(A, B, C) = AB + AB'C + AB'CC'$
 $= AB + AB'(C + C')$
 $= AB + AB'(1) = AB + AB'$
 $= A(B + B') = A(1) = A$
 So, there is no required of NAND gate.
- (c) The maximum and minimum value represented in signed 16 bit 2's complement representations are -32768 and 32767.
- (b)
 $(A + C)(A + AD) + AC + C$
 $= (A + C) \cdot A(1 + D) + AC + C$
 $= (A + C) \cdot A + AC + C$
 $= A \cdot A + AC + AC + C$
 $= A + AC + AC + C$
 $= A + AC + C = A(1 + C) + C = A + C$
- (c) In NAND gate, output is high if any of the input is low. The truth table for the NAND gate is:

A	B	Output
1	1	0
1	0	1
0	1	1
0	0	1

